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Modeling U.S. Supreme Court Briefs with Computational Argumentation

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Abstract. We show how to represent U.S. Supreme Court briefs as a structured argument model that relies entirely on the briefs' existing structure.

Keywords. U.S. Supreme Court, large language model, computational argumentation

Mitigating the public's misunderstanding of how the U.S. Supreme Court makes decision is important [1]. Instead of predicting the outcomes of Supreme Court decisions, we show how to transform the legal arguments presented during the decision-making process into a more digestible form for the public. We focus on the arguments from the Supreme Court briefs, which can influence Supreme Court judges' opinions [2].

Supreme Court briefs, filed after a *writ of certiorari* is granted, can be modeled as a structured argumentation framework because they follow a strict structure [3]. This representation helps the public and stakeholders to understand what arguments were made on their behalf (by the petitioner, respondent, or *amicus curiae*), and how the debate unfolded.

We chose the *Acheson Hotels, LLC v. Laufer*[2] case as an example. Figure 1 shows the table of contents of three sample briefs and how we modeled them: 1) petitioner's brief that was filed before the respondent's brief; 2) the respondent's brief that was filed in response to the petitioner's brief; and 3) one amicus brief that supports the respondent.[3] We model each brief as a structured argument. The subsections under the ARGUMENT section are premises that support the conclusion (content in the CONCLUSION section). Here, we deem each subsection to support the conclusion independently, and the inference from the subsection, as a premise, to the conclusion is defeasible. For example, the part of the respondent's brief we showed in Figure 1 has Sections I and II that support the conclusion, and Section I is supported by Sections I.A, I.B, and I.C.

Attacks are from a later-filed brief to an earlier-filed brief, and the two briefs should not favor the same sides. By keeping the order in which attacks were made, we maintain the decision-making process. Otherwise, it would be counterintuitive for a party to file a brief to attack future arguments. Therefore, arguments in the respondent's brief and the amicus brief in Figure 1 attack the argument in the petitioner's brief. Attacks are classified as undermining (targeting a premise), undercutting (targeting the support

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[2] *Acheson Hotels, LLC v. Laufer*, No. 22–429 (U.S. Dec. 5, 2023).
[3] Petitioner's brief, Respondent's brief, and Amicus brief from *Disability Rights Education & Defense Fund*

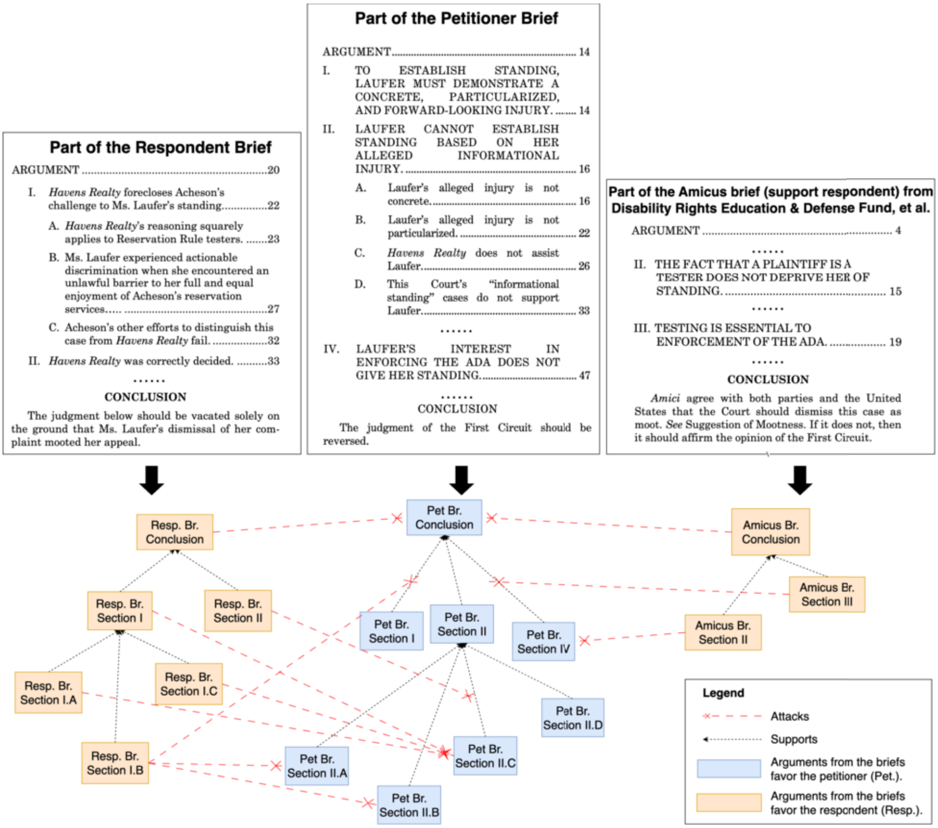

Figure 1. Modeling excerpts from the petitioner brief (middle), the respondent brief (left), and an amicus brief that supports the respondent (right) as a structured argument model

relationship), and rebutting (targeting a conclusion). For example, the respondent's brief concludes that the First Circuit's judgment should be "vacated," which is a rebuttal of the petitioner's brief's conclusion: the judgment should be "reversed."

Constructing the structured argument models can be time-consuming. We explored whether LLMs can assist the construction process by summarizing the sections in the briefs and identifying the attacks between the briefs. LLMs have shown their potential for argument summarization using a benchmarking dataset of U.S. Supreme Court briefs [4]. To see if LLMs can identify attacks between the arguments, we used `Mistral7B` for an initial experiment on five briefs for the *Acheson Hotel* case, and compared the results with Author 1's manual identification of attacks. We chose `Mistral7B` because it performs relation-based argument mining successfully on several non-legal datasets [5]. While a lack of annotated data makes meaningful evaluation challenging, we found the output is still promising.

Our initial attempts support further investigation on combining the strengths of LLMs and computational argumentation to help non-legal experts analyze U.S. Supreme Court briefs. We will investigate how to use LLMs to generate the argument models automatically and use formal semantics, such as ASPIC+ [6], to interpret the briefs further.

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